

Temporally-structured multimodal prediction of relapse risk and survival outcomes in pediatric acute lymphoblastic leukaemia

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Abstract

Acute lymphoblastic leukaemia (ALL) is the most prevalent malignancy in children, and although contemporary treatment achieves five-year survival rates exceeding 90%, approximately 15–20% of patients relapse after apparent remission, with markedly worse prognosis upon re-induction. Existing risk-stratification approaches rely primarily on single-timepoint diagnostic features and fail to integrate the longitudinal nature of clinical data acquisition or to prospectively predict future adverse events. Here we present a temporally-structured multimodal prediction framework applied to a cohort of 301 paediatric ALL patients (2015–2019, Chinese single-centre). We define twelve feature sets ($\mathcal{F}_1 \subseteq \dots \subseteq \mathcal{F}_{12}$) whose incremental composition strictly follows the clinical data-acquisition timeline, from white blood cell count and age at Day 0 to minimal residual disease (MRD) measurements at Day 46. A two-phase system is evaluated: Phase 1 deploys four large language models (LLMs) in a zero-shot setting to probe the upper bound of prediction without training, while Phase 2 trains decision trees and random survival forests on a dynamic case database that grows with annual patient cohorts, explicitly simulating real-world hospital deployment. On the primary risk-stratification task (Final Risk), Phase 2 achieves an AUROC of 0.949 using features available by Day 46, with interpretable decision paths provided by the decision tree. For survival prediction, the best model attains a concordance index of 0.746 and a time-dependent AUC at 24 months of 0.775 for event-free survival (EFS). A retrieval-augmented generation (RAG) case-study component further grounds each prediction in historically similar patients, supporting clinical interpretability. This framework demonstrates that temporally

stratified multimodal learning, aligned with clinical workflows, enables effective prediction at every decision node of paediatric ALL management.

Keywords: paediatric acute lymphoblastic leukaemia, temporal feature design, large language models, risk stratification, survival analysis, retrieval-augmented generation

1 Introduction

Acute lymphoblastic leukaemia (ALL) is the most common childhood malignancy, accounting for approximately 25% of all paediatric cancers [1]. Advances in risk-adapted chemotherapy regimens and supportive care have elevated five-year event-free survival (EFS) rates above 90% in high-income settings; however, the 15–20% of patients who relapse face substantially worse outcomes, with long-term survival dropping to 30–50% following re-induction [2, 3]. Critically, the majority of relapses occur after apparent clinical remission, making prospective identification of high-risk patients a central yet unsolved challenge in paediatric oncology. Current standard-of-care risk stratification (Initial Risk at diagnosis; Final Risk after Day 46 induction assessment) relies on pre-specified clinical cut-offs for age, white blood cell count, cytogenetics, immunophenotype, and MRD, but these categorical rules do not fully exploit the rich multimodal information accumulated progressively during treatment, nor do they yield personalised probability estimates for future relapse or survival.

Computational approaches have been applied to ALL outcome prediction with increasing frequency. Machine learning models trained on flow cytometry profiles [4], gene expression signatures [5], DNA methylation arrays [6], and electronic medical records [7] have demonstrated improved risk separation beyond standard criteria. Nevertheless, two fundamental limitations persist across this literature. First, most methods operate on a *fixed, single-timepoint* feature snapshot at diagnosis, ignoring the structured temporal ordering in which clinical data actually become available—molecular cytogenetics results are routinely unavailable until Day 7–10, and MRD until Day 19–46. A model trained on features pooled from all timepoints may implicitly rely on information that would not be available at the point of clinical decision-making, producing optimistic but non-deployable estimates [9]. Second, standard train-test splits shuffle patients chronologically, whereas in practice a hospital’s knowledge base grows year by year; temporal validation is required to confirm that predictive performance generalises to truly *prospective* patients. Computational flow cytometry approaches have further shown that immunophenotypic data at diagnosis alone may be insufficient for relapse prediction [8], highlighting the need for multimodal and longitudinal integration.

The emergence of large language models (LLMs) with strong zero-shot reasoning capabilities opens a complementary avenue: given structured clinical notes or reports describing a patient’s feature profile, an LLM may distil prognostic patterns encoded in its pre-training without requiring labelled examples from the target cohort [10]. However, systematic evaluations of LLM zero-shot performance on paediatric oncology

outcome prediction—particularly for future events such as relapse and survival—remain absent from the literature. Equally unexplored is whether LLM-based risk scores can serve as survival predictors comparable to purpose-built survival models.

Here we address these gaps with a cohort of 301 Chinese paediatric ALL patients and a *temporally-structured* prediction framework. Our contributions are: (i) a twelve-level feature set hierarchy $\mathcal{F}_1 \subseteq \dots \subseteq \mathcal{F}_{12}$ whose boundaries align exactly with the clinical data-acquisition timeline (Day 0 through Day 46+); (ii) Phase 1, a systematic zero-shot evaluation of four LLMs across all feature levels and six prediction targets, establishing performance baselines without any model training; (iii) Phase 2, a decision tree and random survival forest system trained on a dynamic case database partitioned by patient enrolment year, validating performance under three temporal splits that mimic real-world database growth; (iv) a bone marrow morphology pipeline converting microscopy images to structured text reports via SAM-based cell segmentation and BiomedCLIP clustering, which augments LLM inputs; and (v) a RAG component that retrieves the most similar historical cases to support interpretable clinical narratives for each prediction.

2 Results

2.1 Study cohort and temporally-structured feature design

We assembled a single-centre retrospective cohort of 301 paediatric ALL patients treated at [HOSPITAL NAME] between 2015 and 2019 (Table 1). Median age at diagnosis was [X] years (range [X–X]); 163 (54.2%) patients had B-cell ALL and [X] (XX%) had T-cell ALL. The four primary prediction targets span the treatment timeline: Day 19 MRD positivity ($\text{D19MRD} \geq 1\%$; positive in [X]% of evaluable patients), Day 46 MRD positivity ($\text{D46MRD} \geq 0.01\%$), final risk classification (Final Risk: low-risk [LR] vs intermediate/high-risk [IR/HR], with 159 LR and 142 IR/HR), and relapse occurrence (Yes in 44/301, 14.6%). For survival analysis, EFS events (relapse, induction failure, or death) occurred in 75 patients (24.9%) and overall survival (OS) events (death) in 24 patients (8.0%).

Central to our framework is a twelve-level feature hierarchy $\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \dots \subseteq \mathcal{F}_{12}$ (Table 2), where each level $\mathcal{F}_k$ corresponds to features available at clinical timepoint τ_k . $\mathcal{F}_1$ contains only white blood cell count and age—information present at the moment of hospital admission—while $\mathcal{F}_8$ additionally includes Day 19 and Day 46 MRD values obtained at the end of induction chemotherapy. Feature sets with image-derived inputs ($\mathcal{F}_4, \mathcal{F}_{11}, \mathcal{F}_{12}$) further incorporate bone marrow morphology in the form of either raw microscopy tiles or structured text reports produced by our automated pipeline (Section 4.3). This design ensures that any model trained or evaluated at level k uses only information that would be accessible in clinical practice at timepoint τ_k , avoiding temporal leakage.

Table 1 Cohort characteristics and prediction target distributions.

Characteristic	Value	N (%)
Total patients		301
Enrolment years	2015–2019	
<i>Prediction targets</i>		
Final Risk: LR		159 (52.8%)
Final Risk: IR/HR		142 (47.2%)
Relapse: Yes		44 (14.6%)
D19MRD $\geq 1\%$		[X] ([X]%)
D46MRD $\geq 0.01\%$		[X] ([X]%)
<i>Survival endpoints</i>		
EFS events	relapse+failure+death	75 (24.9%)
OS events	death	24 (8.0%)
<i>Multimodal data availability</i>		
Bone marrow images		[X] (XX%)
Pathology text reports		[X] (XX%)
Morphology features		287 (95.3%)

2.2 Phase 1: LLM zero-shot prediction across the feature timeline

We evaluated four LLMs in a zero-shot setting across all twelve feature levels and four prediction targets, yielding a systematic picture of how predictive performance evolves as more clinical information becomes available (Fig. 2). Models included a proprietary cloud model (Gemini-3.1-Flash-Lite) and three open-weight models deployed locally via vLLM on dual A100 80G GPUs: Gemma-4-31B-it, InternVL3.5-30B-A3B, and Qwen3.6-35B-A3B. Each patient’s feature vector was serialised into natural language and presented to the model with a task-specific system prompt (Methods, Section 4.4); the model output a categorical label or a continuous risk score depending on the task.

Final Risk stratification. At Day 0 ($\mathcal{F}_1$, WBC and age only), Gemini and Gemma already achieved macro-averaged AUROC values of approximately 0.65–0.70 on Final Risk, reflecting the known prognostic importance of WBC count. Performance increased incrementally through $\mathcal{F}_3$ (flow cytometry panel) and $\mathcal{F}_6$ – $\mathcal{F}_7$ (cytogenetics and initial risk classification), with Gemma reaching AUROC = 0.885 at $\mathcal{F}_{10}$ (full Day 7–10 features plus diagnostic report). The addition of MRD values at $\mathcal{F}_8$ produced the largest single improvement across all feature levels, with Gemma achieving AUROC = 0.885 and Gemini AUROC = 0.842 at this level. These results confirm that MRD is the dominant signal for final risk classification, consistent with clinical guidelines.

Relapse prediction. Relapse prediction proved substantially harder for all models and feature levels. Across $\mathcal{F}_1$ – $\mathcal{F}_{12}$, AUROC values for relapse ranged from approximately 0.50 to 0.59 across the four LLMs, with no single feature set or model

Table 2 Feature set design aligned with the clinical data-acquisition timeline.

Exp	Timepoint	Cumulative features added	Targets
$\mathcal{F}_1$	Day 0	WBC, Age	4
$\mathcal{F}_2$	Day 1	+ Blast%	4
$\mathcal{F}_3$	Day 1	+ Immunophenotype + 51 flow cytometry markers	4
$\mathcal{F}_4$	Day 1	Same as $\mathcal{F}_3$ + bone marrow images	4
$\mathcal{F}_{4.1}$	Day 1	Same as $\mathcal{F}_3$ + blast morphology text report	4
$\mathcal{F}_5$	Day 1	$\mathcal{F}_3$ + non-risk clinical factors	4
$\mathcal{F}_6$	Day 5	$\mathcal{F}_3$ + CNS status + Day 5 blast% + Initial Risk	4
$\mathcal{F}_7$	Day 7–10	$\mathcal{F}_6$ + Fusion gene + Karyotype + FISH	4
$\mathcal{F}_8$	Day 46	$\mathcal{F}_7$ + D19MRD + D46MRD	2 [†]
$\mathcal{F}_9$	Day 46	D19MRD + D46MRD only	2 [†]
$\mathcal{F}_{10}$	Day 7–10	$\mathcal{F}_7$ + diagnostic bone marrow text report	4
$\mathcal{F}_{11}$	Day 46+	$\mathcal{F}_8$ + initial + follow-up images & reports	2 [†]
$\mathcal{F}_{12.1}$	Day 7–10	$\mathcal{F}_7$ + blast morphology text report	4

† D19/D46MRD are used as inputs; only Final Risk and Relapse predicted.

achieving consistent improvement. The best results were obtained with morphology-augmented feature sets: Gemma reached AUROC = 0.585 at $\mathcal{F}_{4.1}$ (blast morphology text report added to $\mathcal{F}_3$), and similar improvements of 0.017–0.036 AUROC over raw image inputs were observed across all four models (Fig. 3), indicating that structured morphological statistics convey information about relapse biology beyond what raw tiles provide to LLMs.

MRD prediction. LLMs showed moderate performance on D19MRD and D46MRD prediction (AUROC 0.60–0.80) using pre-MRD features only, suggesting that the models can partially infer early treatment response from the immunophenotypic and cytogenetic profile alone.

Comparative model behaviour. Gemini and Gemma consistently outperformed InternVL and Qwen on most tasks; notably, Qwen3.6-35B-A3B showed near-chance AUROC on D19/D46MRD and Relapse despite competitive performance on Final Risk, suggesting a tendency to predict only the majority class for highly imbalanced tasks. InternVL performed comparably on Final Risk (AUROC $\approx$ 0.58–0.66) but showed weaker text-comprehension of structured feature reports.

2.3 Bone marrow morphology pipeline

Bone marrow smear images were processed by an automated four-stage pipeline: SAM-based cell segmentation, morphological feature extraction (~ 37 features per cell including area, circularity, nucleus-to-cytoplasm ratio, and GLCM texture), Biomed-CLIP zero-shot clustering ($k = 7$ with Hungarian assignment to cell types), and structured text report generation (Methods, Section 4.3). This converts heterogeneous microscopy data into a consistent natural language representation accessible to LLMs.

Comparing $\mathcal{F}_4$ (raw images directly provided to multimodal LLMs) with $\mathcal{F}_{4.1}$ (blast-only morphology text report substituted for images), the text-report approach achieved higher Relapse AUROC for all four models (improvements of +0.017 Gemini, +0.035 Gemma, +0.022 InternVL, +0.036 Qwen), while Final Risk AUROC remained comparable or slightly improved. The all-cell-type report ($\mathcal{F}_{4.2}$) improved Final Risk for Gemma (AUROC 0.854 vs 0.840 for raw images) but was less beneficial for Relapse, suggesting that blast-specific features are more informative for predicting treatment failure. These results demonstrate that domain-specific cell-level characterisation provides predictive signal complementary to global image appearance.

2.4 Phase 2: Temporal validation with a dynamic case database

Phase 2 evaluates whether predictive performance is maintained under a realistic deployment scenario in which a model can only be trained on patients enrolled before the prediction date. We defined three temporal splits based on patient enrolment year τ_i :

$$\mathcal{D}_s^{\text{train}} = \{i : \tau_i \leq \mathcal{T}_s\}, \quad \mathcal{D}_s^{\text{test}} = \{i : \tau_i > \mathcal{T}_s\}, \quad (1)$$

where $\mathcal{T}_1 = 2016$ ($n_{\text{train}} = 99$, $n_{\text{test}} = 201$), $\mathcal{T}_2 = 2017$ ($n_{\text{train}} = 186$, $n_{\text{test}} = 114$), and $\mathcal{T}_3 = 2018$ ($n_{\text{train}} = 283$, $n_{\text{test}} = 17$). All preprocessing statistics (means, standard deviations for z-scoring; one-hot encoding maps) were derived exclusively from $\mathcal{D}_s^{\text{train}}$ and applied to $\mathcal{D}_s^{\text{test}}$, ensuring strict temporal non-leakage. A decision tree classifier (DT; max depth 4, class-weight balanced) and random forest (RF; $n_{\text{estimators}} = 100$) were trained for each feature level and split combination; we report DT as the primary model for interpretability and RF as an ensemble upper bound.

Final Risk — ablation across feature levels (Split 2). Figure 4 shows the DT AUROC for Final Risk across the feature hierarchy on Split 2 ($n_{\text{test}} = 114$, the most statistically robust split). Performance rose from AUROC = 0.810 at $\mathcal{F}_1$ (WBC, age) to AUROC = 0.912 at $\mathcal{F}_7$ (full cytogenetic panel), with the largest gain occurring at $\mathcal{F}_6$ (+0.127 AUROC, reflecting the introduction of Initial Risk and Day 5 peripheral blast percentage into the feature set). Including MRD at $\mathcal{F}_8$ further raised DT AUROC to **0.949** (RF: 0.998), the highest performance across all experimental conditions. The RF AUC approaching 1.0 at $\mathcal{F}_8$ is consistent with the known clinical rule—patients with undetectable MRD at Day 46 are overwhelmingly low-risk—and should be interpreted as a sanity check rather than a novel discovery. The DT, constrained to depth 4, provides a more conservative and clinically transferable estimate.

Temporal learning curve. Across Split 1 ($n_{\text{train}} = 99$) to Split 2 ($n_{\text{train}} = 186$), Final Risk DT AUROC increased from [X] to 0.949, consistent with the intuition that a growing case database enables better discrimination. Split 3 ($n_{\text{test}} = 17$) is underpowered for most tasks and is reported with 95% bootstrap confidence intervals in Supplementary Table S1.

Relapse prediction. Relapse prediction remained challenging for DT and RF alike (best DT AUROC = 0.529, Split 2), mirroring the difficulty observed in Phase 1 and corroborating findings from recent computational flow cytometry studies [8]. This limitation likely reflects the multifactorial biology of relapse—involving treatment

Table 3 Phase 2 classification results on Split 2 ($n_{\text{train}} = 186$, $n_{\text{test}} = 114$). Best DT result per task shown; **bold** = best DT AUROC.

Task	Best Exp	DT AUC	RF AUC	DT F1-macro	DT Bal.Acc.
Final Risk	$\mathcal{F}_8$	0.949	0.998	[X]	[X]
D19MRD	$\mathcal{F}_{4.1}$	0.743	0.807	[X]	[X]
D46MRD	$\mathcal{F}_5$	0.671	0.636	[X]	[X]
Relapse	$\mathcal{F}_1$	0.529	0.465	[X]	[X]

adherence, minimal residual disease kinetics, and late-emerging clonal evolution—that may not be fully captured by the features available in our cohort.

2.5 Survival prediction

We evaluated risk stratification for event-free survival (EFS; 75 events: 45 relapses, 23 induction failures, 7 deaths) and overall survival (OS; 24 deaths) using two complementary approaches. In Phase 1, LLM-derived risk scores (continuous values 0–1 output by the system prompt requesting event probability) were used directly as survival predictors evaluated via Harrell’s concordance index (6) and IPCW time-dependent AUC (7). In Phase 2, a random survival forest (RSF) was trained on temporally partitioned data.

EFS. Phase 1 LLMs achieved peak EFS concordance indices of 0.716 (Gemma, $\mathcal{F}_8$), 0.705 (Gemini, $\mathcal{F}_9$), 0.694 (Qwen, $\mathcal{F}_9$), and 0.682 (InternVL, $\mathcal{F}_9$) across all feature levels. The MRD-only feature set $\mathcal{F}_9$ was consistently optimal or near-optimal for LLM-based EFS prediction, confirming that residual disease burden encodes strong prognostic information accessible to LLMs without any exposure to labelled survival data. Phase 2 RSF on Split 2 attained C-index = **0.746** (time-AUC at 24 months = 0.775, log-rank $p = 0.010$) at $\mathcal{F}_8$, a modest but consistent improvement over the best Phase 1 LLM.

Kaplan-Meier risk stratification. For each model and feature level, patients were dichotomised at the median predicted risk score and Kaplan-Meier curves were estimated for the two strata (Fig. 5). Significant group separation (log-rank $p < 0.05$) was achieved by all four LLMs at $\mathcal{F}_8/\mathcal{F}_9$, and by the Phase 2 RSF at $\mathcal{F}_8$ (Split 2, $p = 0.010$).

OS. With only 24 OS events (8.0% event rate), statistical power for OS models is limited; C-indices ranged from 0.629 to 0.772 across models and feature levels, with no consistent ordering. InternVL reached C-index = 0.772 at $\mathcal{F}_{11}$ (C-index at 24 months AUC = 0.762, $p = 0.0001$); however, results at this feature level incorporate follow-up imaging and should be interpreted with caution regarding temporal availability. Phase 2 RSF Split 3 OS results are omitted due to zero OS events in the 17-patient 2019 test set. A larger cohort or multi-centre study would be required to draw robust conclusions about OS.

Table 4 Survival prediction results. Phase 1: best LLM per model (all patients, no temporal split). Phase 2: Split 2. † p-value from log-rank test.

Phase	Model	EP	Exp	C-idx	AUC@12m	AUC@24m	p [†]
P1	Gemma	EFS	$\mathcal{F}_8$	0.716	[X]	0.771	<0.001
P1	Gemini	EFS	$\mathcal{F}_9$	0.705	[X]	0.777	<0.001
P1	Qwen	EFS	$\mathcal{F}_9$	0.694	[X]	0.752	<0.001
P1	InternVL	EFS	$\mathcal{F}_9$	0.682	[X]	0.737	<0.001
P2	RSF (Split 2)	EFS	$\mathcal{F}_8$	0.746	[X]	0.775	0.010
P1	InternVL	OS	$\mathcal{F}_{11}$	0.772	[X]	0.762	<0.001
P1	Gemma	OS	$\mathcal{F}_8$	0.718	[X]	0.700	<0.001
P2	RSF (Split 2)	OS	$\mathcal{F}_8$	0.629	[X]	0.611	0.489
[PLACEHOLDER: External baseline comparison — to be added]							

2.6 Interpretable prediction via retrieval-augmented generation

To provide clinically grounded explanations for Phase 2 predictions, we implemented a RAG component that supplements each DT prediction with a set of the $K = 5$ most similar historical cases retrieved from the training database, followed by an LLM-generated narrative synthesising the decision-tree path and the retrieved precedents (Methods, Section 4.6). Case retrieval used cosine similarity in the same z-scored feature space as the DT:

$$\mathcal{N}(i) = \arg \text{top-}K_{j \in \mathcal{D}_s^{\text{train}}} \frac{\mathbf{z}_i \cdot \mathbf{z}_j}{\|\mathbf{z}_i\|_2 \|\mathbf{z}_j\|_2}, \quad K = 5. \quad (2)$$

Figure 6 illustrates two representative cases from the $\mathcal{F}_8$ /Split 3 experiment. In Case 1, a patient correctly predicted as IR by the DT (path: $\text{WBC} > 34.5 \rightarrow \text{D46MRD} > 0.01\% \rightarrow \text{CD10+} = \text{true} \rightarrow \text{IR}$), the top-5 retrieved historical cases all shared persistently elevated MRD and cytogenetic abnormalities, and the LLM narrative coherently attributed the high-risk classification to MRD non-clearance corroborated by historical precedent. In Case 2, a borderline patient whose prediction changed between Split 1 and Split 2 models, the narrative highlighted conflicting signals in the retrieved cases (similar MRD but divergent cytogenetic profiles), suggesting areas where additional investigation—such as next-generation sequencing—might resolve ambiguity. These examples demonstrate that the RAG component can produce clinically meaningful explanations that go beyond the rule statement of the decision tree alone.

3 Discussion

This study presents a temporally-structured multimodal framework for paediatric ALL outcome prediction that explicitly mirrors the clinical data-acquisition workflow, enabling meaningful predictions at each decision node from admission to end-of-induction. Several key findings warrant discussion.

MRD as the dominant predictor. The most striking result across both Phase 1 and Phase 2 is the predictive jump attributable to MRD values: DT AUROC for

Final Risk increased from 0.912 ($\mathcal{F}_7$, cytogenetics) to 0.949 ($\mathcal{F}_8$, plus D19/D46 MRD) under Split 2 conditions. Similarly, LLM EFS C-index peaked at feature levels that included MRD. This quantitatively confirms the central role of MRD in the clinical risk-stratification guideline [11], and provides an AUROC estimate that directly characterises the discriminative value of adding MRD to cytogenetic profiling. From a clinical translation perspective, the DT achieves 0.949 AUROC for Final Risk with a transparent depth-4 decision tree that can be directly audited by clinicians, which is important for institutional adoption [12].

Temporal validation and database growth. The consistent improvement in Final Risk AUROC as the training database grows from 99 (Split 1) to 186 (Split 2) patients suggests that accumulating historical cases continues to benefit predictive performance, motivating a prospective case-database design for clinical AI systems. The smallest test set (Split 3, $n = 17$) also yielded the highest nominal DT AUROC (0.979), likely an artefact of the small and possibly unrepresentative 2019 cohort rather than genuine improvement, underscoring the importance of interpreting results from small test sets with bootstrap confidence intervals.

LLM zero-shot as a competitive baseline. Phase 1 LLMs reached Final Risk AUROC values of 0.842–0.885 without any labelled training data, a result competitive with the Phase 2 DT ($\leq \mathcal{F}_7$ conditions). This indicates that the clinical knowledge encoded during LLM pre-training includes a meaningful understanding of paediatric ALL risk factors. LLMs may therefore be particularly valuable in settings where historical case databases are small or absent—for example, at the time a new treatment protocol is introduced—serving as a strong prior until sufficient labelled data accumulate.

The challenge of relapse prediction. Relapse prediction proved intractable for both LLMs (AUROC ≈ 0.50 – 0.59) and DT/RF (best DT AUROC = 0.529), consistent with the broader literature [4, 8]. Our feature set does not include genomic or epigenomic data (e.g., copy-number variation, DNA methylation), which have shown modest incremental utility in single-centre studies [6]. Future work integrating multi-omics data within the temporal feature hierarchy may improve relapse prediction.

Limitations. This is a single-centre retrospective study; external validation in independent cohorts is required before clinical deployment. The relatively small cohort ($n = 301$) limits the statistical power for OS analysis (24 events) and for the smallest temporal split ($n = 17$). All Phase 1 LLM experiments used zero-shot prompting; fine-tuning on labelled clinical notes from this or similar cohorts may yield further improvements. External baseline comparisons against published methods trained on independent datasets have not yet been performed and are a planned extension of this work.

4 Methods

4.1 Dataset and ethics

The cohort comprised 301 patients diagnosed with acute lymphoblastic leukaemia at [HOSPITAL NAME] between 2015 and 2019. All patients were treated with risk-stratified chemotherapy protocols consistent with contemporary Chinese paediatric oncology guidelines. Ethics approval was obtained from [ETHICS NUMBER]; patient data were de-identified prior to analysis. For each patient the dataset includes: (i) structured clinical data (demographics, haematological indices, immunophenotype, 51-marker flow cytometry panel, cytogenetics, MRD values); (ii) bone marrow aspirate microscopy images (Giemsa-stained smear, ~ 20 fields per patient, total [X] images); (iii) diagnostic and follow-up bone marrow pathology text reports.

4.2 Temporally-structured feature design

We define feature sets as a strictly nested sequence

$$\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \dots \subseteq \mathcal{F}_{12}, \quad (3)$$

where each $\mathcal{F}_k$ contains all features clinically available at timepoint τ_k . Boundaries were established by consulting the standard ALL treatment protocol: $\mathcal{F}_1$ (Day 0): WBC count and age; $\mathcal{F}_2$ (Day 1): adds blast percentage from bone marrow aspirate; $\mathcal{F}_3$ (Day 1): adds immunophenotype classification and 51 flow cytometry surface markers (42 CD antigens and 9 other markers including cMPO, TdT, and HLADR); $\mathcal{F}_6$ (Day 5): adds CNS involvement status, Day 5 peripheral blast percentage, and Initial Risk classification; $\mathcal{F}_7$ (Day 7–10): adds fusion gene, karyotype, and FISH results; $\mathcal{F}_8$ (Day 46): adds D19MRD and D46MRD numerical values. Image-based feature sets $\mathcal{F}_4$, $\mathcal{F}_{4.1}$, $\mathcal{F}_{4.2}$, $\mathcal{F}_{11}$, and $\mathcal{F}_{12.x}$ incorporate bone marrow morphology as described in Section 4.3.

For each patient i , the input vector at level k is $\mathbf{x}_i^{(k)} \in \mathbb{R}^{|\mathcal{F}_k|}$. Categorical features (Immunophenotype, Fusion Gene) were encoded by natural-language description for LLM inputs, and by one-hot encoding for DT/RF inputs with mappings derived exclusively from the training split (Section 4.5).

4.3 Bone marrow morphology pipeline

Each bone marrow aspirate image was processed through a four-stage automated pipeline.

Stage 1 — Cell segmentation. We applied SAM 3.1 [13] with a text prompt (“*complete dark-stained nucleated cell*”) to each image after colour-based pre-filtering (HSV purple range 120–160°) to restrict prompt sampling to haematoxylin-stained regions. Segmented cell patches were filtered by darkness ratio (< 0.82) and minimum stained pixel fraction (> 0.05).

Stage 2 — Morphological feature extraction. For each retained cell, ~ 37 features were computed from the segmentation mask and corresponding RGB patch, including geometric descriptors (area, perimeter, circularity, eccentricity, solidity,

major/minor axis, aspect ratio), textural descriptors (GLCM contrast, correlation, energy at multiple angles), nucleus-to-cytoplasm (N/C) ratio, nucleoli count, and vacuole percentage.

Stage 3 — Cell-type classification. BiomedCLIP [14] was applied in a zero-shot clustering mode with $k = 7$ prototype classes (blast, granulocyte, megakaryocyte, monocyte, plasma cell, red cell, other). Hungarian algorithm assignment mapped cluster indices to cell-type labels using cosine similarity to BiomedCLIP text embeddings.

Stage 4 — Report generation. Per-patient statistics (cell counts, mean morphological features per class, blast proportion) were aggregated across images and serialised into a structured natural-language report supplied as additional context to LLM prompts in $\mathcal{F}_{4.1}$, $\mathcal{F}_{4.2}$, $\mathcal{F}_{10}$, $\mathcal{F}_{12.1}$.

4.4 Phase 1: LLM zero-shot inference

Four LLMs were evaluated. Gemini-3.1-Flash-Lite was accessed via the Google AI API. Gemma-4-31B-it, InternVL3.5-30B-A3B, and Qwen3.6-35B-A3B were deployed locally with vLLM [15] using tensor parallelism across two A100 80G GPUs.

For classification tasks (D19MRD, D46MRD, Final Risk, Relapse), each patient’s features were serialised into a key–value list and submitted with a task-specific system prompt instructing the model to output a bracketed label (e.g., `[[Positive]]` / `[[Negative]]`). For survival risk scoring, the model output a continuous probability (0.0–1.0) of an adverse event, parsed by extracting the first numerical token and clipping to $[0, 1]$. All experiments used temperature = 0 (greedy decoding) to ensure reproducibility. Per-patient outputs were cached; the full cohort of 301 patients was scored for each combination of feature level, task, and model, yielding $12 \times 4 \times 4 = 192$ experimental cells for classification and $12 \times 2 \times 4 = 96$ for survival scoring.

4.5 Phase 2: Decision tree and random forest with temporal splits

For each temporal split s (Eq. 1) and feature level k , numerical features were z-score normalised using training-set statistics only:

$$z_{ij}^{(\text{test})} = \frac{x_{ij}^{(\text{test})} - \mu_j^{(\text{train})}}{\sigma_j^{(\text{train})}}, \quad j = 1, \dots, |\mathcal{F}_k|. \quad (4)$$

Categorical features used one-hot encoding with maps derived from the training set. Missing values were imputed with the training-set median per feature.

A decision tree classifier was trained with `max_depth = 4`, `class_weight = ‘balanced’`, `criterion = ‘gini’`, and `random_state = 42`. The Gini impurity at node t is

$$G(t) = 1 - \sum_{k=1}^K \hat{p}_{tk}^2, \quad (5)$$

where $\hat{p}_{tk}$ is the empirical class proportion. A random forest with $n_{\text{estimators}} = 100$ and `class_weight` = ‘balanced’ was trained in parallel as an ensemble upper bound; RF results are reported in supplementary tables.

Primary evaluation metric is AUROC; F1-macro and balanced accuracy are reported as secondary metrics. Bootstrap 95% confidence intervals (1000 resamples) are reported for Split 3 ($n_{\text{test}} = 17$).

4.6 Phase 2: RAG retrieval and LLM narration

For a test patient i , the $K = 5$ most similar training cases were retrieved by cosine similarity in the training-standardised feature space (Eq. 2). An LLM prompt was then constructed containing: (i) the patient’s feature values, (ii) the DT-predicted label and probability with the human-readable decision path (split conditions from root to leaf), and (iii) the structured profiles of the top-5 retrieved cases including their actual outcomes. The model was instructed to produce a 3–5 sentence clinical narrative explaining the prediction based on the decision path and the historical precedents. LLM narratives are produced by Gemini-3.1-Flash-Lite and are used exclusively for qualitative case-study presentation; they do not contribute to any quantitative evaluation.

4.7 Survival analysis

EFS was defined as time (months) from diagnosis to first of: relapse, induction failure, or death, with censoring at last follow-up. OS was defined as time from diagnosis to death, censored at last follow-up.

Phase 1 continuous risk scores $\hat{r}_i \in [0, 1]$ were treated as survival predictors. Harrell’s concordance index is

$$\hat{C} = \frac{\sum_{i \neq j} \mathbf{1}[T_i < T_j] \mathbf{1}[\delta_i = 1] \mathbf{1}[\hat{r}_i > \hat{r}_j]}{\sum_{i \neq j} \mathbf{1}[T_i < T_j] \mathbf{1}[\delta_i = 1]}, \quad (6)$$

where T_i is follow-up time, $\delta_i \in \{0, 1\}$ is the event indicator.

Time-dependent AUC at time t^* was computed using inverse probability of censoring weighting (IPCW) [16]:

$$\widehat{\text{AUC}}(t^*) = \frac{\sum_{i: T_i \leq t^*, \delta_i = 1} \sum_{j: T_j > t^*} \mathbf{1}[\hat{r}_i > \hat{r}_j] \hat{w}_i}{\sum_{i: T_i \leq t^*, \delta_i = 1} \sum_{j: T_j > t^*} \hat{w}_i}, \quad (7)$$

where $\hat{w}_i = 1/\hat{G}(T_i)^2$ and $\hat{G}(\cdot)$ is the Kaplan-Meier estimator of the censoring distribution. Time points evaluated: $t^* \in \{12, 24, 36\}$ months.

Kaplan-Meier curves were estimated for high- and low-risk groups (dichotomised at the median predicted risk score):

$$\hat{S}(t) = \prod_{t_i \leq t} \left(1 - \frac{d_i}{n_i}\right), \quad (8)$$

with group separation assessed by the log-rank test.

Phase 2 survival used a random survival forest (RSF; [17]; $n_{\text{trees}} = 100$) with the same temporal train/test splits as the classification experiments; evaluation used identical C-index and IPCW time-AUC metrics.

4.8 Statistical analysis

AUROC was computed with the `scikit-learn` library [18]; C-index and IPCW AUC with `scikit-survival` [19]. No multiple testing correction was applied given the exploratory nature of this study; all reported p -values are two-sided.

Data availability

The clinical dataset used in this study contains patient health information and cannot be made publicly available due to privacy regulations. De-identified data may be available to qualified researchers through a data access agreement with [HOSPITAL NAME]; requests should be directed to the corresponding author.

Code availability

Analysis code, LLM prompts, and the bone marrow morphology pipeline will be made available at https://github.com/YanbeiJiang/ALL_prediction_NCS upon acceptance.

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Author contributions

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Competing interests

The authors declare no competing interests.

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Fig. 1 Overview of the temporally-structured multimodal prediction framework. **a**, The three parallel timelines defining the framework: the feature acquisition timeline (Day 0 to Day 46+, $\mathcal{F}_1$ through $\mathcal{F}_{12}$), the prediction target timeline (D19MRD, D46MRD, Final Risk, Relapse, EFS, OS), and the dynamic case database timeline (patient enrolment year 2015–2019 defining temporal splits). **b**, Phase 1 architecture: four LLMs receive serialised patient features and output class labels (classification tasks) or risk scores (survival tasks) without task-specific training. **c**, Phase 2 architecture: a decision tree trained on historically accumulated cases generates a prediction and decision path; the RAG module retrieves the $K = 5$ most similar historical cases; an LLM synthesises a clinical narrative.

Fig. 2 Phase 1 feature ablation: AUROC vs feature level. **a**, Final Risk. **b**, Relapse. **c**, D19MRD. **d**, D46MRD. Each panel shows the macro-averaged AUROC across the twelve feature levels for four LLMs (Gemini, Gemma, InternVL, Qwen). Vertical dashed lines indicate clinically important timepoints: Day 1 (flow cytometry), Day 5 (early response), Day 10 (cytogenetics), Day 46 (MRD; $\mathcal{F}_8$).

Fig. 3 Bone marrow morphology pipeline and its predictive contribution. **a**, Pipeline schematic: SAM cell segmentation $\rightarrow$ morphological feature extraction $\rightarrow$ BiomedCLIP cell-type clustering $\rightarrow$ structured text report generation. **b**, Comparison of LLM AUROC for Relapse prediction across three image-handling strategies: raw images ($\mathcal{F}_4$), blast-only text report ($\mathcal{F}_{4.1}$), and all-cell-type text report ($\mathcal{F}_{4.2}$) for four LLMs. **c**, Same comparison for Final Risk AUROC.

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Fig. 4 Phase 2 decision tree feature ablation and temporal learning curve. **a**, Final Risk DT AUROC across feature levels $\mathcal{F}_1$ – $\mathcal{F}_{12.1}$ for three temporal splits (Split 1: 2015–16 train; Split 2: 2015–17 train; Split 3: 2015–18 train). Shaded band: 95% bootstrap CI for Split 3. **b**, Training set size vs AUROC learning curve for Final Risk (DT and RF) and Relapse (DT and RF) at $\mathcal{F}_8$.

Fig. 5 Survival prediction: Kaplan-Meier curves and performance summary. **a**, EFS Kaplan-Meier curves (high vs low risk, median split) for the best Phase 1 model (Gemma, $\mathcal{F}_8$, all patients). **b**, EFS KM curves for Phase 2 RSF (Split 2, $\mathcal{F}_8$). **c**, C-index heatmap: all LLMs $\times$ all feature levels for EFS. **d**, Same for OS. Number at risk and shaded 95% CI shown in panels a–b.

Fig. 6 RAG case study. Two representative cases from the $\mathcal{F}_8$ /Split 3 experiment. For each case: patient feature summary, DT decision path (root to leaf), top-3 retrieved similar historical cases with outcomes, and LLM-generated clinical narrative. Case 1: correctly classified IR patient with persistent MRD. Case 2: borderline case with conflicting signals between MRD clearance and cytogenetic risk.

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